Diagnosis and Identification of Exceptional Learners

Learners with disabilities, the gifted and the twice-exceptional



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Chapter 1

1 Concepts of diagnosis and identification

Screening, identification, diagnosis and eligibility are four terms often used as if they were synonyms, yet they are in fact four distinct steps, each with its own question, its own tools and its own responsible party. In this chapter we sketch the conceptual map on which all the chapters of the book are organized, and we place it within its regulatory framework in Saudi Arabia.

Scene from the teachers' lounge

At the start of the school year, Munira, a teacher, received the files of the students in her third-grade class. In Abdullah's file was a hospital report stating that he had been "diagnosed with autism spectrum disorder." Sultan's file held the result of a reading screening the school had conducted the previous month, marked "needs follow-up." Reem's file was empty, but last year's teacher had left Munira a note: "Keep an eye on Reem. I think she's gifted."

> Munira wonders: Is Abdullah entitled to special education services because he has a diagnosis? Does Sultan have a learning disability because the screening flagged him? And is one teacher's opinion enough to consider Reem gifted?

The answer to all three questions is the same: "Not necessarily." Why this is so becomes clear only once we draw careful distinctions between steps that are frequently confused in practice. That is the work of this chapter, and we will return to the three students at its close.

Chapter objectives

After studying this chapter, the reader should be able to:

1. Determine the purpose of assessment in a given situation and the type of tool suited to it.
2. Distinguish among screening, identification, diagnosis and eligibility determination by the question each step asks and the kind of evidence it requires.
3. Compare the three models of disability in terms of the diagnostic question toward which each one points.
4. Map the special education categories in Saudi regulations onto the categories of US law and onto medical classification systems.
5. Identify which principle of fair assessment has been violated in a given assessment situation.
6. Build a **RIOT/ICEL** matrix for gathering evidence about a referred student, indicating the source of each piece of information in it.
7. Plan a conversation with a family about a teacher's concern, following a structured sequence of steps.

8 Detect common errors in a given assessment situation and propose ways to avoid them.

9 Judge whether the available evidence is sufficient to move from one step of the assessment pathway to the next, and justify that judgment.

Introduction

Every school year, teachers and assessment and diagnosis teams make decisions that bear directly on the paths of particular students: who is referred for assessment, who meets the criteria for a given category, and who needs special education services or a gifted program. Many of the errors in these decisions stem less from the limitations of the tools than from confusing distinct steps that go by similar names. A screening result is treated as a diagnosis, a medical diagnosis is taken as an educational eligibility decision, and a single teacher's opinion is deemed sufficient to judge a student gifted, just as Munira finds in the opening scene. For this reason we open the book by pinning down concepts before turning to tools and categories, since it is the question each step asks that determines the kind of evidence required, who is responsible for it, and how much of it is enough to move on to the next step.

We begin the chapter with the purposes of assessment, starting from the premise that assessment is a means of making decisions (Section 1.1). We then distinguish among screening, identification, diagnosis and eligibility determination (Section 1.2), and among measurement, testing and assessment (Section 1.3). Next we present the models of disability and the question toward which each points (Section 1.4), followed by medical and educational classification systems and the regulatory framework in Saudi Arabia (Section 1.5). We then turn to the principles of fair assessment (Section 1.6), the **RIOT/ICEL** framework for organizing sources of evidence (Section 1.7), and the roles of the team and the family (Section 1.8), and we close with the general assessment pathway and its common errors (Section 1.9).

1.1 Why do we assess? The purposes of assessment

Assessment is a means of making a decision. The first question an assessor asks, therefore, is "What decision do I need to make?" Only after that comes the question "Which test should I give?" A tool that is excellent for one decision may be useless for another. Specialists in educational measurement identify several purposes of assessment, each with its own question (Salvia et al., 2017):

Table 1.1: Purposes of assessment, the question each purpose answers, and an example of a tool for it.

Purpose	Question it answers	Example of a tool
Screening	Who needs a closer look?	A short screening checklist given to all kindergarten children
Diagnosis and classification	Does the student have a disability, a disorder or giftedness? Of what kind?	A comprehensive battery: intelligence, achievement, adaptive behavior and observation
Eligibility determination and placement	Is the student entitled to special services? In what setting should they be provided?	A team decision based on the diagnosis and its impact on learning
Instructional planning	What should we teach? Where should we start?	Criterion-referenced tests (which compare a student's performance with a defined level of mastery rather than with the performance of peers), and error analysis
Progress monitoring	Is the student learning? Is the plan working?	Brief, repeated measures such as curriculum-based measurement (**Curriculum-Based Measurement, CBM**)

Purpose	Question it answers	Example of a tool
Program evaluation	Is the program meeting its goals for its students as a whole?	Aggregated data on the program's students across years

In this book we focus on the first two purposes, with some attention to the third. The teacher, however, is present in all of them, because diagnostic information must ultimately feed into instructional planning; otherwise it loses its value.

1.2 Screening, identification, diagnosis and eligibility

Picture these four steps as a funnel: at the top are all students, and at the bottom are those who receive special services. The funnel narrows at each layer, because each step asks a more precise question than the one before it and requires stronger evidence.

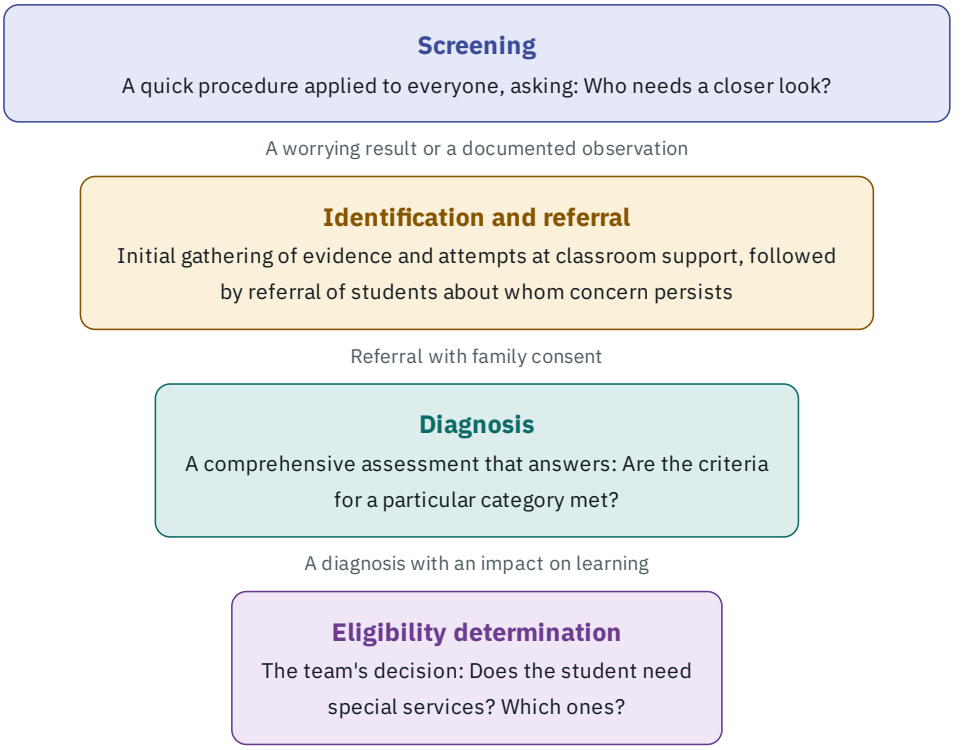


Figure 1.1: From screening to eligibility. The number of students shrinks at each layer, while the depth of the questions and the strength of the evidence required increase.

Screening

Screening is applied to an entire group, such as all first graders, using a short tool that does not require a specialist. Its aim is to ensure that no child who needs a closer look slips past us. Screening tools are therefore preferably high in **sensitivity** (**Sensitivity**), that is, able to capture the largest possible proportion of those who actually have the problem, even if they also flag some who do not, because the cost of missing a child is greater than the cost of examining a child who turns out not to need it.

Sensitivity alone, however, is not enough to judge a screening tool. The tool also needs acceptable **specificity** (**Specificity**), meaning the ability to rule out those who do not have the problem; otherwise it will swamp the team with

unnecessary referrals. A school principal will care about a third question as well: if a student's result is negative, how likely is it that the student truly does not have the problem? This is the **negative predictive value** (**Negative predictive value**), the most important question to ask of a screening, because it measures how confident we can be that we have not missed anyone. Its magnitude, like that of the positive predictive value, depends jointly on the properties of the tool and on the prevalence of the problem in the group screened (Chapter 2 works through these concepts with numbers in Section 2.6).

Misconception: "The screening flagged him, so the problem is confirmed"

A positive screening result is an invitation to look more closely; diagnosis is a later stage that requires broader evidence. Screening tools are deliberately designed to catch more than they should, and when a problem is rare in a group, many of those flagged by the screening may in fact be free of it. The reverse holds as well: a negative result is no certificate of health. If the teacher's or the family's concern persists, a documented observation is reason enough to move on to the identification stage.

Identification and referral

After screening, or after a teacher's observation, the identification stage (**Identification**) begins: the stage of gathering initial evidence. At this stage the teacher and the school try to provide support within the classroom, whether by adjusting the method of instruction or by offering additional small-group teaching, while monitoring the student's response. If the difficulty persists despite this support, the student is referred for a comprehensive assessment. Referral is a significant decision, because it signals that the school has tried what it can and that this has not been enough.

In the field of giftedness, this stage is usually called **nomination**. Teachers and families nominate students in whom they see indicators of giftedness, and those students are then assessed. Relying on nomination alone, however, misses many gifted students, because nominators tend to notice those who display their excellence clearly in achievement and behavior, and overlook those whose giftedness is hidden behind a second language, a disability or a quiet disposition (Chapter 12 addresses nomination in detail).

The concept of identification: Why is it different in the field of giftedness?

In this book we restrict "identification" to the stage that precedes referral, a terminological choice that gives each step of the pathway a name of its own. In the literature, however, the term **Identification** usually denotes the entire process of determination, up to and including the decision, and in the Arabic literature it overlaps with "**al-kashf**" (detection), as in the name of the "National Program for Gifted Identification," which encompasses both testing and classification. Yet this broad sense does not carry the same content in the two fields. In the field of giftedness, identification is the complete process of determining who needs gifted services, and it is an educational process; the term "diagnosis" is avoided here because giftedness is neither a disorder nor a deficit whose cause must be sought. The process starts from the program's definition and goals, draws on multiple criteria and on local norms that compare students with peers in their own setting, and links the tool to the type of service offered: a measure of mathematical reasoning suits an enrichment program in mathematics but is not, on its own, appropriate for a program in creative writing. Its outcome is a decision to place a student in a program or to provide a service, one that is tied to the student's educational need and open to revision whenever the evidence or the program changes (National Association for Gifted Children [NAGC], 2019).

In the field of disabilities, by contrast, "identification" usually means determining eligibility for special education services under the categories recognized by the education system, as in the US Individuals with Disabilities Education Act and its implementing regulations published in the Code of Federal Regulations (IDEA, 2004; 34 C.F.R. §300.8). For some categories, such as autism spectrum disorder and attention-deficit/hyperactivity disorder, this educational decision is paralleled or preceded by a clinical diagnosis under the Diagnostic and Statistical

Manual of Mental Disorders (American Psychiatric Association, 2022) or the International Classification of Diseases (ICD-11). The educational decision thus meets the medical one without either replacing the other, as Section 1.5 explains in detail.

The two pathways converge in twice-exceptional learners: the same student may need a clinical diagnosis of a disability and an educational identification of giftedness, and each pathway may obscure what the other reveals if the team pursues one without the other (Chapter 13 examines this overlap in detail). So when you encounter either term in another source, infer its meaning from the context and the field, and do not assume that it refers to the narrow stage we intend here.

Diagnosis

Diagnosis answers the question: Does the student meet the criteria for a particular category? It requires a comprehensive team assessment using multiple tools and multiple sources. A diagnosis may be medical, issued by a physician or a qualified psychologist according to a diagnostic manual, or educational, issued by a school team according to education regulations.

Eligibility determination

Here we arrive at the distinction that puzzled Munira. Diagnosis answers the question “What does the student have?”, whereas eligibility answers the question “Does the student need special services in order to learn?” A student may have a medical diagnosis yet learn successfully in the general classroom without needing special services, and so not be eligible for them. Conversely, a student may need services without yet having a medical diagnosis, in which case the student is assessed educationally.

The equation that governs eligibility

In most education systems, eligibility for special education services requires two conditions together: the student must meet the criteria for one of the disability categories, and the disability must affect the student's learning to the point that he or she needs specially designed instruction. If either condition is absent, the student is not eligible, even if the other is met. Saudi regulations phrase the second condition in their own words: an impairment "to the extent that it necessitates the provision of special education services" (Ministry of Education [Al-Ma'arif], 1422 AH [Arabic]).

1.3 Measurement, testing and assessment

These words overlap in everyday use, but each has a specific meaning in the literature, and knowing the difference helps in reading reports and sources accurately:

Table 1.2: The difference among testing, measurement, assessment and evaluation.

Term	Meaning	Example
Test	An organized set of tasks or questions administered in a specified way	An intelligence test, or a 20-word spelling test
Measurement	Expressing a characteristic in numbers according to rules	Scored 14 out of 20, or has a standard score of 85
Assessment	The complete process of gathering information from multiple sources in order to make a decision	Tests, interviews, observation and records, ending in a decision

Term	Meaning	Example
Evaluation	Making a judgment about the worth or adequacy of something; in some systems also used to mean the comprehensive assessment of a student	Evaluating a program's effectiveness, or the comprehensive evaluation of a student's eligibility

The key point here is that testing is a small part of assessment. Sound assessment combines tests with observations, interviews and a review of records and work samples, as we will see in the evidence-gathering framework in Section 1.7.

1.4 Models of disability

Before we diagnose a disability, we must ask: What is disability in the first place? Where does the problem lie? How we answer these two questions is a stance with consequences, for it determines what we measure and where we look for solutions. There are three main models:

The medical model

The problem lies in the individual

Disability is a defect in the individual's body or functions, to be diagnosed and treated or compensated for. The focus is on the impairment and its causes.

In diagnosis: looking for symptoms, matching them against criteria,

The social model

The problem lies in the environment

Disability arises from the barriers society places in front of those who differ: buildings a wheelchair cannot reach, curricula that ignore individual differences. The focus is on removing barriers (Oliver, 1990).

The biopsychosocial model

The problem lies in the interaction between the two

Disability results from an interaction between the individual's health condition and the environmental and personal factors surrounding him or

and determining severity.

In diagnosis: asking about the barriers in the student's environment and what needs to change around the student.

her (Engel, 1977). This model is the foundation of the International Classification of Functioning, Disability and Health issued by the World Health Organization (World Health Organization [WHO], 2001).

In diagnosis: describing the impairment, what the student is able to do, and the environmental factors that hinder or help.

It is often said that modern diagnosis follows the third model, but this claim is a generalization that needs qualifying. The Diagnostic and Statistical Manual of Mental Disorders, fifth edition, text revision (DSM-5-TR), remains at its core a categorical classification belonging to the medical model: the criteria are either met or not met, even though the manual adds severity specifiers and levels of support that move it closer to graded description (American Psychiatric Association, 2022). The biopsychosocial model, by contrast, underlies the International Classification of Functioning specifically, a descriptive classification that addresses functioning, participation and environmental factors and leaves the diagnosis of conditions to the medical systems. There is also growing discussion in the literature of dimensional models that describe psychological difficulties along graded continua rather than in discrete categories, one of the best known being the Hierarchical Taxonomy of Psychopathology (**HiTOP**).

Good practice comes closest to combining the two logics: the assessor judges whether criteria are met using a categorical logic when the system requires it, and then describes the impact of the condition and its relationship to the environment using a biopsychosocial logic. When we say that a student with autism spectrum disorder "requires substantial support," we are describing the student's relationship to the environment as well as what the student has. The same student may need less support in a calm, structured classroom and more in a crowded, changeable one.

1.5 Classification systems

Diagnosis rests on classification systems that define categories and their criteria. The most important thing for a teacher to know is that there are two kinds of systems, medical and educational, and that each serves a different purpose.

Table 1.3: Medical and educational classification systems and their purposes.

System	Issuing body	Purpose	Who uses it
Diagnostic and Statistical Manual of Mental Disorders (DSM-5-TR)	American Psychiatric Association, 2022	Diagnosing mental and developmental disorders using detailed criteria	Psychiatrists and psychologists

System	Issuing body	Purpose	Who uses it
International Classification of Diseases, Eleventh Revision (ICD-11)	World Health Organization; adopted in 2019 and in force since January 1, 2022; its clinical descriptions and diagnostic requirements for mental and developmental disorders were published in 2024	Classifying all diseases and health conditions, including developmental ones, for diagnostic and statistical purposes	Health systems around the world
International Classification of Functioning, Disability and Health (ICF)	World Health Organization, 2001	Describing what a person can do and participate in, and the environmental factors at play, rather than diagnosing the condition	Rehabilitation and service-planning teams
Educational categories	Each country's education legislation and regulations	Determining who is entitled to special education services	School teams and education departments

Clinical diagnosis relies on one of the first two manuals (American Psychiatric Association, 2022; WHO, 2019, 2024). The two converge for most developmental categories, though they differ in naming and detail, differences taken up in the category chapters of Part Three.

An example of an educational system: the US categories

The US Individuals with Disabilities Education Act (**Individuals with Disabilities Education Act, IDEA**) was reauthorized in 2004 as the Individuals with Disabilities Education Improvement Act (**IDEIA**), though it is still known by its original acronym. The law specifies thirteen categories eligible for services: autism, deaf-blindness, deafness, emotional disturbance, hearing impairment, intellectual disability, multiple disabilities, orthopedic impairment, other health impairment, specific learning disability, speech or language impairment, traumatic brain injury (resulting from a blow or impact), and visual impairment including blindness. The law also permits a developmental delay category for children aged 3 to 9 (IDEA, 2004).

Note three important points about this list:

1. **Attention-deficit/hyperactivity disorder (Attention-Deficit/Hyperactivity Disorder, ADHD) is not a separate category.** A student diagnosed with it usually qualifies for services under the category of "other health impairment." In other words, medical diagnoses and educational categories do not always coincide. Saudi regulations depart from this arrangement, as we will see in the next section.
2. **Giftedness is absent.** The law concerns disabilities, and gifted services in the United States are governed by separate legislation. This explains why a twice-exceptional student, that is, a gifted student who also has a disability or disorder, falls between two systems that are not coordinated with each other (Chapter 13 discusses this group in detail).
3. **The labels differ from those of the medical manual.** "Emotional disturbance" is a broad educational category that encompasses conditions bearing different names in the medical manual.

1.5.1 The regulatory framework in Saudi Arabia

In this section we present the Saudi education system in the same detail as the US example, since it is the framework within which the assessment procedures discussed in this book are carried out. Special education in Saudi Arabia is organized at two levels: general laws that establish the rights of persons with disabilities, and regulatory documents issued by the Ministry of Education that define the categories, eligibility criteria and assessment and diagnosis procedures. The identification of gifted students is organized along a pathway separate from both. Table 1.4 summarizes the principal documents.

Table 1.4: The principal regulatory documents for diagnosis and identification in Saudi Arabia.

Document or body	Issuance	What matters to the assessor
Regulations for Special Education Institutes and Programs	Ministry of Education [Al-Ma'arif] (the former name of the Ministry of Education), Decision No. 1674 of 5/4/1422 AH (Ministry of Education [Al-Ma'arif], 1422 AH [Arabic])	Definitions of the categories and admission requirements for each; a section on assessment and diagnosis (Articles 78–82) setting out its rules, its team and its seven steps; the components of the individualized education program (Articles 83–85); and the obligations of the institute or program toward the family (Article 76)

Document or body	Issuance	What matters to the assessor
Special Education Organizational Guide and Special Education Procedural Guide	Ministry of Education, first edition for the 1436–1437 AH school year, which counts the Regulations among its foundations (Ministry of Education, 1436 AH-a, 1436 AH-b [Arabic])	The recognized categories and the eligibility criteria for each, the composition of the assessment and diagnosis team, parental consent, timelines for reevaluation, and eligibility for early intervention
Law on the Rights of Persons with Disabilities	Royal Decree No. (M/27) of 11/2/1445 AH (2023), in 33 articles; it replaced the Disability Welfare Law issued by Royal Decree No. (M/37) of 23/9/1421 AH	The right of persons with disabilities to "modern assessment and diagnostic tools" and to early intervention programs (Article 8), and to screening and diagnostic services and the "early detection of disability" (Article 9)
Authority for the Care of Persons with Disabilities	Its organizational statute was approved by the Council of Ministers in 1439 AH (2018) (Al-Jazirah, 2018 [Arabic])	A financially and administratively independent body whose board is chaired by the Minister of Human Resources and Social Development; it works to safeguard the rights of persons with disabilities, develop policies, programs and services, and support research (Saudipedia, n.d. [Arabic])

Document or body	Issuance	What matters to the assessor
National Program for Gifted Identification, and the regulations for gifted programs at the Ministry of Education	The program was launched in 2011 as a partnership among Mawhiba, the Ministry of Education and the National Center for Assessment (King Abdulaziz and His Companions Foundation for Giftedness and Creativity [Mawhiba], n.d. [Arabic]); the General Administration for the Gifted organizes gifted classes under a dedicated guide (Ministry of Education, n.d. [Arabic])	The Mawhiba Multiple Mental Abilities Test, the nomination criteria for gifted classes, and the rule for combining them

Note that the two education documents use the term ʿawq (impairment) where this book uses iʿāqa (disability). The Regulations use "mental retardation," dividing it into "educable" and "trainable" according to ranges of IQ scores, whereas the Organizational Guide uses "mental impairment" and additionally requires deficits in at least two adaptive domains. In both cases the condition meant is what this book calls "intellectual disability."

Categories and their criteria: a comparison with the US system

The Organizational Guide recognizes eleven categories. Table 1.5 maps them onto the categories of US law, together with the main eligibility criteria the Guide specifies.

Table 1.5: The categories of US law (IDEA) and their counterparts in the Special Education Organizational Guide, with the main eligibility criteria.

US legal category	Counterpart in the Organizational Guide	Main eligibility criteria in Saudi regulations

US legal category	Counterpart in the Organizational Guide	Main eligibility criteria in Saudi regulations
Intellectual disability	Mental impairment	IQ no higher than 75 on the Wechsler scale or 69 on the Stanford-Binet Fifth Edition, or the equivalent of either, together with deficits in at least two adaptive skill domains. US law sets no numerical cut-off score
Specific learning disabilities	Learning disabilities	A marked discrepancy between ability and achievement in one academic area, or a disorder in one of the basic psychological processes; with intellectual and sensory impairments and behavioral disorder ruled out; and evidence that general education services are inappropriate or of limited effectiveness. The text specifies no numerical size for the discrepancy and makes no mention of the response to intervention model. US law, by contrast, prohibits states from requiring a discrepancy and obliges them to permit the response to intervention model (34 C.F.R. §300.307)
Deafness; hearing impairment	Hearing impairment (deaf and hard of hearing)	A hearing loss of 70 decibels or more for deafness, and between 35 and 69 decibels for hard of hearing, in the better ear with hearing aids. The US definition is functional, with no numerical threshold
Visual impairment including blindness	Visual impairment (blind and low vision)	Low vision: acuity between 6/24 and 6/60 meters. Blindness in the Regulations: acuity below 6/60 in the better eye after correction, or a visual field of less than 20 degrees

US legal category	Counterpart in the Organizational Guide	Main eligibility criteria in Saudi regulations
Deaf-blindness	Deaf-blind	A separate category in both systems
Multiple disabilities	Multiple impairments	A separate category in both systems
Orthopedic impairment; other health impairment	Physical and health impairment	A single category in the Guide corresponding to two categories in US law
Autism	Autism spectrum disorder	The Guide defines it by persistent deficits in social communication and social interaction across multiple contexts, in wording that matches the first criterion of the diagnostic manual
Speech or language impairment	Language and speech disorders	A separate category in both systems
Emotional disturbance	Behavioral and emotional disorders	A separate category in both systems; the Saudi label is broader in its wording
Not a separate category (usually served under "other health impairment")	Attention-deficit and hyperactivity disorders	"Hyperactivity, attention deficit or both, requiring intervention according to an accredited report"

US legal category	Counterpart in the Organizational Guide	Main eligibility criteria in Saudi regulations
Traumatic brain injury	No separate category in the Guide	We found no separate category for it in the Organizational Guide
Developmental delay (ages 3–9)	Early intervention: developmentally delayed children and children at risk (up to age 6)	A delay of two standard deviations below the mean in two or more domains, or of 25%; or exposure to at least three environmental risk factors
Giftedness: outside the law	Giftedness: outside the Special Education Organizational Guide	Organized along a separate pathway (the National Program for Gifted Identification and the programs of the General Administration for the Gifted)

Sources: Ministry of Education (1436 AH-a [Arabic]); Ministry of Education [Al-Ma'arif] (1422 AH [Arabic]) for the definition of blindness; and US law and its implementing regulations (IDEA, 2004; 34 C.F.R. §300.8). The Guide also sets general eligibility requirements, among them the statutory age for admission to first grade, with special education students permitted an exception of no more than three years.

Identification of gifted students

The identification of gifted students in Saudi Arabia rests on two complementary pathways. The first is the "National Program for Gifted Identification," which targets students from third grade through the first year of secondary school at three levels. Students may sit the test once at each level, and the program administers the "Mawhiba Multiple Mental Abilities Test" upon registration by the family (Mawhiba, n.d. [Arabic]). The second consists of the Ministry of Education's gifted classes, which draw their students from those classified as gifted by the national program. If places remain unfilled,

the classes complete their numbers using four criteria: a mental abilities test, personal traits, academic achievement and creative products. A student must satisfy two or more criteria, one of which must be the mental abilities test with a score of at least 120 (Ministry of Education, n.d. [Arabic]). For a recent overview of gifted education in Saudi Arabia, see Alfaiz and colleagues (Alfaiz et al., 2022).

In the Saudi context: What the comparison reveals

- **Attention-deficit/hyperactivity disorder is a separate category in Saudi Arabia.** This brings the educational category closer to the medical diagnosis than the US system does, but it also makes the "accredited report" a cornerstone of eligibility, which raises the stakes for the quality of the assessment on which that report was based.
- **The learning disabilities model in the text rests on discrepancy, exclusion and nonresponse to general instruction.** The text specifies neither the size of the discrepancy nor how nonresponse is to be established. Recent studies indicate that practices resembling the response to intervention model exist in Saudi schools, but that they are largely informal and lack systematic progress monitoring, and that the absence of formal adoption of the model is among the barriers to its implementation (Almulla & Mahama, 2026; Altheiabi et al., 2025).
- **Gifted students with disabilities fall between two documents.** The Special Education Organizational Guide does not address gifted students, and the gifted guides we reviewed do not address students with disabilities. We found no published regulatory text that either permits or prohibits combining a gifted program with learning disabilities services. A recent exploratory study estimated that about 5.9% of a sample of 6,875 students in four private schools in Riyadh may be twice-exceptional, and called for policies specific to them (Almulhim et al., 2026).

The documents cited in this section were consulted in September 2026, and we found no published edition of the Organizational and Procedural Guides more recent than their first edition; readers should consult whichever edition is in force at the time of reading. **Question for**

reflection: If you were invited to join a committee updating the Organizational Guide, which of these three gaps would you address first, and why?

1.6 Principles of fair assessment

An assessment that ends in a diagnosis changes the course of a student's life: the school the student attends, the curriculum the student learns, how others see the student, and how the student sees himself or herself. For this reason, legislation and professional standards have set out principles that protect students from error and injustice. Among the clearest formulations are those in the implementing regulations of US law governing evaluation procedures (34 C.F.R. §300.304). The Standards for Educational and Psychological Testing treat fairness as a third foundation of testing alongside validity and reliability, devoting their third chapter to it (American Educational Research Association [AERA] et al., 2014). Table 1.6 sets these principles against the provisions of Saudi regulations.

Table 1.6: Principles of fair assessment, what they mean in practice, and their counterparts in Saudi regulations.

Principle	What it means in practice	Counterpart in Saudi regulations
Multiple tools and strategies	No single measure is used as the sole basis for any decision; a single score, however precise, is not enough on its own to support a judgment.	"Not relying solely on the results of a single measure when determining a pupil's eligibility," and using formal and informal methods (Rules, Article 80; Organizational Guide)

Principle	What it means in practice	Counterpart in Saudi regulations
Assessing all areas of suspected need	If there is doubt about hearing, language or attention, each is assessed, rather than only the area for which the student was referred.	A multidisciplinary team and information gathered from varied sources (Rules, Article 80)
Appropriate language and mode	The student is assessed in his or her native language or mode of communication; a deaf student is assessed in sign language if that is the student's language.	We found no explicit provision in either document specifying the language or mode of assessment
Freedom from bias	Tools are chosen so that they do not disadvantage the student because of culture or background.	Standardized tests that are “valid, reliable and suited to the child's environment” (Rules, Article 80)
Measuring what is intended to be measured	If a student has a motor disability, his or her intelligence is not measured with a test that depends on motor speed, lest the score fall because of the student's hand rather than ability.	“Appropriate measures and clear instructions” (Rules, Article 80), with no detail on adapting administration for students with sensory and motor disabilities
Validity of the tool and qualifications of the examiner	The tool is used for the purpose for which its validity has been established, and it is administered by a trained person according to its instructions.	Using “tools and measures approved by the Ministry of Education,” administered and interpreted “by qualified specialists” (Organizational Guide). We found no published text specifying the qualification levels required for each class of standardized psychological test

Principle	What it means in practice	Counterpart in Saudi regulations
Informed family consent	The family knows what will happen and why, and consents in writing before the process begins.	Written consent from the parent or guardian after formal notification; if no reply is received within fourteen days, this is deemed consent (Organizational Guide; the Regulations specify two weeks)
Confidentiality	Only those entitled to access the information may see it, and it is stored in a way that protects it.	"Maintaining the confidentiality of all information" (Rules, Articles 76 and 80); the Personal Data Protection Law (1443 AH) classifies health data as sensitive data

Sources: implementing regulations of US law (34 C.F.R. §300.304), Standards for Educational and Psychological Testing (AERA et al., 2014), Ministry of Education [Al-Ma'arif] (1422 AH [Arabic]), and Ministry of Education (1436 AH-a [Arabic]).

The comparison reveals the strength of Saudi regulations on the principles of multiple tools and examiner qualifications, and their silence on the language and mode of assessment, a silence with real consequences for the assessment of deaf students and of students whose native language is not Arabic. The consent rule deserves particular attention. In the professional literature, informed consent is an affirmative act that the family performs after it understands, and silence does not count as consent under the US definition (34 C.F.R. §300.9). Treating a lack of response as consent may streamline procedures, but it does not ensure that the family understood what it agreed to. The team would therefore do well, even when the rule allows it to proceed, to seek explicit consent and to arrange a meeting in which it explains what will take place.

These principles do not take the place of the ethical codes of the professions in which assessors work, such as the principles of the US National Association of School Psychologists (National Association of School Psychologists, 2020).

Technical standards set out how a test is built and used; ethical codes set out what the assessor owes the student and the family (Chapter 2 discusses the difference between the two in detail).

1.7 Sources of evidence: the RIOT/ICEL framework

How can we be sure that the evidence we have gathered is sufficient and varied? School psychology offers a simple, practical framework for this purpose, known by its acronym (**RIOT/ICEL**; Christ, 2008). The acronym is formed from the initial letters of its words: the first part (**RIOT**) covers the methods of gathering information (**Review, Interview, Observe, Test**), and the second part (**ICEL**) covers the domains (**Instruction, Curriculum, Environment, Learner**). The framework is a two-dimensional matrix: **four methods** of gathering information, applied across **four domains** in which the problem may lie.

- **The four methods:** review, interview, observation and testing.
- **The four domains:** instruction, curriculum, environment and the learner.

The central idea of the framework is that the problem may lie outside the student: in the way he is taught, in a curriculum pitched above his level, or in a noisy environment. An assessment that looks at the student alone may diagnose a disability that does not exist. Table 1.7 presents the matrix with examples in each cell.

Table 1.7: The RIOT/ICEL matrix, with examples in each cell. A balanced assessment fills cells in every row and every column rather than confining itself to the testing cell in the learner row.

Instruction **Instruction**	Lesson plans and the teaching methods used so far	Asking the teacher how the skill was presented	Observing the clarity of explanation and the pace of teaching	Trying a different teaching method and measuring the response to it

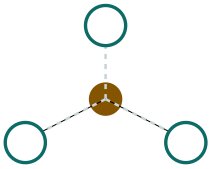
Curriculum Curriculum	Difficulty level of the textbook and the sequence of skills	Asking the teacher about the prerequisites for the skill	Observing whether the task matches the student's level	Measuring text readability, or assessing prerequisite skills
Environment Environment	Attendance records, and changes of school and teacher	Asking the family about circumstances at home	Observing noise, lighting and where the student sits	Classroom environment scales
Learner Learner	Medical reports, previous results and samples of the student's work	Asking the student what he finds difficult	Observing his behavior across different situations	Ability and achievement tests and diagnostic scales

For the teacher

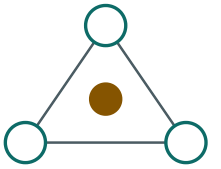
Most of the cells in this matrix are in your hands before they are in the specialist's. You are the one who knows how the student is taught, the level of the curriculum and the classroom environment, and you hold the student's daily work. When you attach this information to the referral, you give the team half the picture before it begins.

1.8 The team and the family

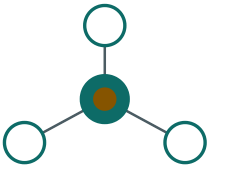
No single specialist has all the knowledge needed to understand a student fully. Diagnosis is therefore the work of a team, which, depending on the case, may include the psychologist, the special education teacher, the classroom teacher, the speech-language pathologist, the physician, the occupational therapist, the audiologist, the social worker and the family. Teams differ in how they work together:



Multidisciplinary
Each specialist assesses alone and writes a separate report. Their recommendations may conflict without their ever meeting.



Interdisciplinary
Each specialist assesses within his or her own field; they then meet, share their findings and reach a joint decision and plan.



Transdisciplinary
The team assesses the child together in a single session, with one member working directly with the child under the guidance of the others. This model is common in early childhood.

Figure 1.2: Three models of teamwork. The solid circle at the center represents the child, the hollow circles around it represent the specialists, and the lines represent communication: dashed between each specialist and the child alone, solid between specialists when they work together.

The assessment team in Saudi regulations

The Regulations assign assessment and diagnosis to a team made up of the relevant special education teacher, the behavioral training teacher (the psychologist), the child's guardian, and anyone whose participation the case requires (Article 81). Among the psychologist's duties they list “conducting assessment and diagnosis” using formal tools, such as standardized intelligence scales and adaptive behavior scales, and informal tools, such as interviews, observation and checklists (Articles 48 and 49) (Ministry of Education [Al-Ma'arif], 1422 AH [Arabic]). The Organizational Guide widens the team to include the school principal, the vice principal for student affairs, the special education teacher, the general education teacher, the student counselor or social worker, specialists according to the disability category, the physician or nurse where available, and the student's guardian “if possible” (Ministry of Education, 1436 AH-a [Arabic]). The phrase “if possible” is weaker

than what the Regulations themselves establish when they make the guardian a member of the team, and weaker than what this book holds in the next section.

The family as a partner in assessment and decision-making

The family has known the student since birth, and in situations no one else sees. Its role in assessment goes well beyond answering questions:

- **The holder of consent:** Assessment does not begin without the family's informed consent, and the family has the right to ask questions and to object.
- **A source of developmental history:** When the child walked, when he began to talk, and what changed and when.
- **A witness to behavior in a different setting:** What the family sees at home may differ from what the teacher sees, and the difference is itself information.
- **A partner in the decision and the plan:** The family attends the results meeting, understands the findings and takes part in setting goals.

The Regulations grant the family explicit rights in this regard: the guardian's permission must be obtained for diagnosis, for preparing the program and for approving any modification; the family is to be invited to take part in diagnosis, program planning and evaluation; their child's condition is to be explained to them “in understandable language”; and the student is to be rediagnosed at the family's request if they believe he has not been placed in the appropriate program (Article 76). The Regulations also require the family's participation in preparing, implementing, evaluating and monitoring the individualized education program (Ministry of Education [Al-Ma'arif], 1422 AH [Arabic]).

For the teacher: When you talk to the family about your concern

1. Begin with what the student does well; the family needs to know that you see their child as a whole person.
2. Describe what you have noticed in terms of specific behaviors and examples, and avoid labels: say “He doesn't turn when I call his name,” not “I think he has autism.”
3. Ask: “Do you notice anything similar at home?” Then listen.
4. Explain the next step and the reason for it: “I suggest that a specialist team assess him, so that we can understand how to help him.”
5. Do not promise an outcome, and do not suggest a diagnosis to the family; that is outside your role, and it may be wrong.

1.9 The general pathway and its common errors

Detailed procedures vary across countries and categories, but most systems follow a broadly similar pathway:

1 **Observation and screening**

The teacher or the family notices indicators, or the student is flagged in screening results.

Teacher, family, screening team

2 **Classroom support and documentation**

Instructional adjustments and additional support for a set period, with the response measured and documented.

Teacher, supported by the school team

3 **Referral and consent**

If the response is insufficient, a referral is written with the evidence attached, and the family's consent is requested.

Teacher, school administration, family

4 **Comprehensive assessment**

The team gathers evidence from multiple sources, using tools suited to each area of suspected need.

Multidisciplinary team

5 **Decision meeting**

The team and the family discuss the results and determine whether the criteria are met and whether the student is eligible for services.

Team, family

6 **Plan, monitoring and reevaluation**

The individualized education program (**Individualized Education Program, IEP**) is developed, progress is monitored, and the student is reevaluated periodically or as needed.

Special education teacher, classroom teacher, family

Some systems set binding timelines for these steps. In the US system, for example, the initial evaluation must be completed within sixty days of parental consent, unless the state sets a different timeline (IDEA, 2004). The purpose of such timelines is to ensure that a student does not wait for months without services or a decision. Table 1.8 sets each step against the provisions of the two systems.

Table 1.8: Steps of the general pathway in the US system and their counterparts in Saudi regulations.

Step	In the US system (as an example)	Counterpart in the Saudi system

Step	In the US system (as an example)	Counterpart in the Saudi system
1. Observation and screening	The state is obliged to locate, identify and evaluate children with disabilities (**Child find**; 34 C.F.R. §300.111)	A right to the "early detection of disability" (Law on the Rights of Persons with Disabilities, Article 9); for giftedness: the National Program for Gifted Identification, upon registration
2. Classroom support and documentation	For learning disabilities, evidence of appropriate instruction and repeated measurement of progress is required, and the response to intervention model may be used (34 C.F.R. §300.307, §300.309)	For learning disabilities, it must be established that general education services are inappropriate or of limited effectiveness, without specifying how this is to be established; the student may be placed under observation "for a full semester" where needed (Regulations, Article 80)
3. Referral and consent	Informed written parental consent before the initial evaluation (34 C.F.R. §300.300)	Written consent from the guardian after formal notification; no response within 14 days is treated as consent (Organizational Guide)
4. Comprehensive assessment	Completed within sixty days of consent, unless the state sets a different timeline	Carried out by the assessment team using tools approved by the Ministry, in seven steps that begin with notifying the guardian and end with the final report (Regulations, Article 82). We found no timeline in either document for completing the initial evaluation that corresponds to the sixty-day limit

Step	In the US system (as an example)	Counterpart in the Saudi system
5. Decision meeting	A team that includes the parents determines eligibility, and the family receives a copy of the evaluation report (34 C.F.R. §300.306)	The assessment team issues the final report; the guardian is a member of the team under the Regulations, and "if possible" under the Guide
6. Plan and reevaluation	An individualized education program with specified components (34 C.F.R. §300.320); reevaluation at least once every three years, and not more than once a year unless the family and the agency agree (34 C.F.R. §300.303)	An individualized education program covering present levels of performance, long-term and short-term goals written in measurable behavioral terms, services, and performance criteria and dates for evaluating the plan (Regulations, Articles 83–85); rediagnosis no sooner than six months and no later than eighteen months after the initial diagnosis, and thereafter as needed (Organizational Guide); the family may request rediagnosis (Regulations, Article 76)

Sources: Ministry of Education [Al-Ma'arif] (1422 AH [Arabic]), Ministry of Education (1436 AH-a [Arabic]), the Law on the Rights of Persons with Disabilities (1445 AH [Arabic]), and the implementing regulations of US law (34 C.F.R. Part 300).

Common errors in diagnosis

Table 1.9: Common errors in diagnosis, how they occur and how to avoid them.

Error	How it occurs	How to avoid it

Error	How it occurs	How to avoid it
Diagnostic overshadowing	New symptoms are attributed to a disability the student is already known to have, so that another disorder is missed; for example, anxiety in a student with an intellectual disability is explained away as "part of his disability" (Reiss et al., 1982).	Asking every time: Does the known disability really explain this symptom?
Confirmation bias	The assessor starts from a hypothesis, sees the evidence that supports it and overlooks the evidence against it. A teacher convinced that a very active student has attention-deficit/hyperactivity disorder, for example, records every moment of distraction but fails to notice his sustained attention when a task captures his interest.	Deliberately seeking alternative explanations and attending to discrepancies among sources.
Reliance on a single score	A decision is made because the score fell above or below the cut-off score (the threshold that the scale or the system sets for the decision), without regard to the confidence interval, the range within which the student's true score is likely to fall, or to other evidence.	The principle of multiple tools, and reading the score within its surrounding confidence interval (Chapter 2 explains both concepts in detail).
Overdiagnosis or underdiagnosis	A group of students is represented among those diagnosed at a higher or lower rate than expected because of language, sex or social and cultural background.	Fair tools, local norms and periodic review of referral data.
Overlooking the effect of labeling	The team forgets that a diagnosis can change what people expect of the student, so that less is asked of him.	Having the report describe strengths and focus on the support the student needs.

Back to Ms. Munira's classroom

- **Abdullah:** He has a medical diagnosis of autism spectrum disorder, but his eligibility for services is determined by the impact of the disorder on his learning. He may need extensive services, or only simple classroom accommodations, that is, changes in how he learns or is tested that leave the level of the curriculum untouched. The medical report in his file is evidence the assessment team can draw on in reaching the eligibility decision.
- **Sultan:** The screening flagged him, which means he needs a closer look; concluding that he has a learning disability would be premature. The next step is reading support and monitoring of his response, before any referral.
- **Reem:** The teacher's opinion is a useful nomination and the starting point for identification. Reem needs an assessment based on multiple criteria, that is, on evidence from more than one source: an ability test, a creativity measure, teacher ratings and samples of her work. In Saudi Arabia she could start along either of two pathways: registering for the National Program for Gifted Identification when she reaches a grade it targets, or the nomination criteria for the Ministry of Education's gifted classes.

A running student file: Sultan's file at the outset

Sultan's file can serve as a running exercise, with each chapter of Parts One and Two adding what is relevant to it: scores and confidence intervals, results from formal tools, repeated measurement data, and then the application of an identification model. Here is what the file contains now, organized by the **RIOT/ICEL** framework:

Table 1.10: Sultan's file at the outset: what we have and what is missing.

Cell	What we have	What is missing
Review / Learner	Reading screening result: "Needs monitoring"	Detailed screening scores, results from previous years, and samples of his writing and reading
Review / Environment	Nothing	Attendance record, and whether he has moved between schools or teachers
Interview / Environment	Nothing	Asking the family about reading at home, and about hearing, vision and developmental history
Observation / Instruction and curriculum	Nothing	Observing the reading lesson: Are the texts at his level? Is decoding taught explicitly?
Testing / Learner	Nothing yet	Brief, repeated measures of reading fluency during the weeks of support

The question that remains open is this: Is his weakness in reading a response to instruction that did not suit him, or a difficulty that persists despite appropriate instruction? It can only be answered after weeks of documented support.

Conclusion

The argument we have developed in this chapter holds that assessment begins with the decision to be made and then selects its tool, and that each step in the pathway has its own question: screening sorts, identification gathers initial evidence and refers, diagnosis matches criteria, and eligibility determines the need for services. It follows that no step should be asked for what it cannot provide. A screening result does not establish a disability, a medical diagnosis does not automatically confer educational eligibility, and a teacher's opinion is the beginning of identification. It follows, too, that evidence should be gathered from multiple tools and sources, that instruction, curriculum and environment should be examined before a shortfall is attributed to the student, and that the family should be a partner in the decision as well as a source of information.

Some of what this chapter has presented, however, is a matter of terminology or of judgment. Using “identification” for the stage before referral is a terminological choice we have adopted for this book; the meaning of the term differs between the field of giftedness and the field of disabilities; medical and educational categories do not always coincide; and combining criterion-based judgment with a description of functioning in context calls for professional judgment that no ready-made rule can supply. Deciding whether the evidence is sufficient to move from one step to the next remains a professional judgment, one that must be reasoned and that weighs the numbers, the documents and all the other evidence.

1.10 Summary, glossary and activities

Chapter summary

- Assessment is a means to a decision, and its first question is: What decision do we want to make?

- Screening sorts, identification gathers initial evidence and refers, diagnosis matches criteria, and eligibility determines the need for services. Using “identification” for the stage before referral is a terminological choice made for this book.
- A screening tool is judged by its sensitivity and its specificity together, and its negative predictive value tells us how confident we can be that we have not missed anyone.
- Medical diagnosis and educational eligibility are two different questions, and medical and educational categories do not always coincide.
- The Diagnostic and Statistical Manual is categorical at its core, the biopsychosocial model underpins the International Classification of Functioning, and good practice combines criterion-based judgment with a description of functioning in context.
- In Saudi Arabia, diagnosis is governed by the Regulations for Special Education Institutes and Programs (1422 AH) and the Special Education Organizational Guide (1436–1437 AH); the Law on the Rights of Persons with Disabilities (1445 AH) establishes the right to diagnosis and early detection; and the identification of gifted students follows a separate pathway.
- Fair assessment draws on multiple tools and sources, is conducted in the student's language, uses tools valid for their purpose, and rests on the family's informed consent.
- The **RIOT/ICEL** framework is a reminder that the problem may lie in instruction, curriculum or the environment, as well as in the student.

Glossary of terms

Screening	A quick procedure applied to an entire group to determine who needs a more in-depth assessment.
Identification	In this book: the stage of gathering initial evidence and attempting support within the classroom, in preparation for a referral decision. In much of the literature, the term denotes the entire process of determination up to the decision; in the field of giftedness, it is a multi-criteria educational process for determining who needs gifted services, not a clinical diagnosis.
Identification; detection	A term that overlaps in the Arabic literature with “identification” and is generally used for the process of locating and identifying gifted students or students with disabilities, as in the “National Program for Gifted Identification” and the “early detection of disability.”
Referral	A formal request for a comprehensive assessment of the student, accompanied by the evidence and the family's consent.
Nomination	A proposal by a teacher, the family or the student himself that a student be assessed within a gifted identification program.
Diagnosis	A judgment that the criteria for a particular category apply to the student, based on a comprehensive assessment.

Eligibility	A decision on whether the student is entitled to special services, based on the diagnosis and its impact on his learning.
Sensitivity	The ability of a tool to capture those who actually have the problem; a property strongly required of screening tools.
Specificity	The ability of a tool to rule out those who do not have the problem.
Negative predictive value	The proportion of those with a negative result who are actually free of the problem; it is affected by the prevalence of the problem in the group.
Informed consent	The family's agreement after learning what will be done, why, and who will have access to the results.
Diagnostic overshadowing	Attributing new symptoms to a known disability, which leads to another condition being missed.
Confirmation bias	The assessor's tendency to see the evidence that supports a prior hypothesis and to overlook the evidence against it.
Cut-off score	The threshold that a scale or a system sets for making a decision.
Multidisciplinary team	A group of specialists from different fields who take part in assessing the student, each usually working alone.

Interdisciplinary team	A team in which each member assesses the student within his or her own field; members then share their findings and reach a joint decision and plan together.
Transdisciplinary team	A team that assesses the child together in a single session, with one member working directly with the child under the guidance of the others.
Biopsychosocial model	A model that views disability as the outcome of an interaction between the individual's health condition and environmental and personal factors.
Review, Interview, Observe, Test / Instruction, Curriculum, Environment, Learner	A matrix for gathering evidence through four methods (review, interview, observation and testing) across four domains (instruction, curriculum, environment and the learner).

Review questions

1. Identify the purpose of the assessment and the appropriate type of tool in each situation: (a) a school wants to know which of all its first graders need monitoring in reading; (b) a teacher wants to know whether the plan for a student with a difficulty in arithmetic is working; (c) an education directorate wants to know whether its gifted program is meeting its objectives.
2. What is the difference between diagnosis and eligibility? Give an example of a student who has a diagnosis but is not eligible for services.
3. Why is it preferable for screening tools to be sensitive, even if they sometimes flag students who do not need help? And why is a school

principal more concerned with a tool's negative predictive value than with its positive predictive value?

4. This book uses "identification" for the stage before referral, while the "National Program for Gifted Identification" uses the word "al-kashf" (detection). What is the difference between the two meanings? And why does this difference matter to readers of the literature in other languages?
5. How does the diagnostic question in the Diagnostic and Statistical Manual (**DSM-5-TR**) differ from that in the International Classification of Functioning (**ICF**)? Which of the two expresses the biopsychosocial model more clearly?
6. Under which category is a student diagnosed with attention-deficit/hyperactivity disorder served under US law, and under which in the Saudi Organizational Guide? What corresponds in the Guide to the category of "traumatic brain injury"? And where does giftedness fall in each system?
7. A student with low vision obtained a low score on an intelligence test that relies on small pictures. Which principle of fair assessment was violated? Can you find a provision in Saudi regulations that protects the student from this error?
8. The Organizational Guide provides that a guardian's failure to respond within 14 days is treated as consent. Discuss this provision in light of the principle of informed consent, and propose a practice that safeguards the family's rights without stalling the process.
9. Fill in the cells of the **RIOT/ICEL** matrix for a second grader suspected of having a reading difficulty, and indicate the source of each piece of information.
10. What is diagnostic overshadowing? How might it delay the detection of attention-deficit/hyperactivity disorder in a student diagnosed with a learning disability?
11. **Case for application:** Ghada is thirteen and in the first year of intermediate school. Her mother has brought the school a report from a private clinic stating that she has "attention-deficit/hyperactivity disorder,

predominantly inattentive presentation"; the report was based on an interview with the mother and a rating scale completed by the mother alone. The mother is asking for Ghada to be admitted to the program for attention-deficit/hyperactivity disorders. Her mathematics teacher says that Ghada forgets her homework and does not finish tests on time, while her science teacher sees no problem. Her grades were excellent throughout elementary school and have dropped this year in mathematics alone. She has received no classroom adjustments, and her hearing and vision have not been checked since she started school. (a) At which step of the general pathway does Ghada actually stand? (b) Is the available evidence sufficient for an eligibility decision under the Organizational Guide? Justify your answer. (c) Identify three cells of the **RIOT/ICEL** matrix that need evidence before any decision, and rank them by priority. (d) Which of the common errors in Table 1.9 might the team fall into here? And what question would you ask the mother about when the difficulties began?

Table 1.11: Alignment of chapter objectives with review questions and learning activities.

Objective	Assessed by
1. Identify the purpose of assessment and the type of tool	Question 1
2. Distinguish among the four steps of the pathway	Questions 2, 3 and 4
3. Compare models of understanding disability	Question 5
4. Map Saudi categories onto US and medical categories	Question 6 and Activity 1
5. Identify which principle of fair assessment has been violated	Questions 7 and 8
6. Build a RIOT/ICEL matrix	Question 9 and Activity 2
7. Frame the conversation with the family	Activity 3

Objective	Assessed by
8. Detect common errors	Question 10 and part (d) of Question 11
9. Judge the sufficiency of evidence	Question 11 (Case for application)

Learning activities

General caution: Any activity involving real students or families is carried out with the consent of the family and the school administration, without any data that could identify the student, and under the supervision of a specialist if it involves administering a tool. Nothing gathered for training purposes may serve as the basis for a decision, and its results are communicated only through official channels.

1. **Mapping the local pathway:** Draw a student's pathway from observation to the plan under Saudi regulations, using Tables 1.5 and 1.8. Then identify three issues that the documents reviewed in this chapter either do not address or address only in general terms, and propose an alternative regulatory wording for each. If possible, ask a special education teacher about actual practice in his or her school on these issues, without mentioning any student. *Output:* A one-page diagram and a comparative paragraph. *Approximate time:* Two hours.
2. **A matrix for a case:** Fill in the **RIOT/ICEL** matrix for Ghada's case in Question 11, or for a student you know, with his or her identity concealed and under the conditions set out in the general caution. Which cells do you have information for? Which need someone to fill them in, and who is that? *Output:* A completed matrix with a column for the source of each piece of information. *Approximate time:* 45 minutes.
3. **A conversation with a family (paired activity):** One partner plays the teacher and the other the guardian; the teacher practices communicating a concern using the five steps in Section 1.8, and then the partners switch roles. *Output:* Written notes from each partner on the other's performance at each step. *Approximate time:* 30 minutes.

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This book traces the stages of diagnosis and identification systematically, from referral and the gathering of evidence to the interpretation of that evidence and the decisions built on it. It goes beyond defining the categories and setting out their criteria to examine the nature of measurement instruments, the procedures for administering them, the qualifications required of those who administer them, the principles for interpreting their results, and the point at which the evidence is sufficient to support a judgment.

The book brings together two fields that the literature usually treats separately: disabilities and disorders, and giftedness and twice-exceptionality. It reads international standards and the American experience in light of published Saudi regulations, and describes the status of the Arabic version of each instrument it presents.

Its fifteen chapters address the foundations of measurement; formal, informal and alternative methods of gathering evidence; and identification models; then take up the categories of exceptionality one by one, supported by decision pathways, instrument profiles and applied cases. They close with fairness in diagnosis, report writing, and the development of the individualized education program.

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